

YOUR OFE 2025 REPORT

AUTHOR
Farmers DataLab

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2025 Cropping Season

Field 3 | NRate 57 lb N/ac | **Research question:** Does a cover crop increase corn nitrogen status and biomass nitrogen concentration?

Welcome to your 2025 farm report. This report summarizes the on-farm nitrogen experiment conducted in your field, where cover crop strips (treatment) were compared against no-cover-crop controls.



Figure 1. Field 3 — treatment layout. ■ Treatment ■ Control

Take-home message: Across all four nitrogen indicators (NBI, Chlorophyll, Biomass %N, and CSNT) corn plants in the cover crop strips showed better nitrogen status than those in the control strips. Leaf readings were greener and more nitrogen-sufficient, plant tissue had noticeably more nitrogen, and end-of-season stalk nitrate revealed a clear difference between treatments: control stalks were well below the sufficiency threshold, while cover crop stalks were within the optimal range. Keep in mind that the tissue and stalk results come from very few samples, so they are early signals rather than proof. Your soil health score is solid overall, but building biological activity through continued cover cropping is the clearest path to unlocking even more nitrogen from the soil over time.

Results Summary

Metric	Control	Treatment	Difference	p-value
 NBI (Nitrogen Balance Index)	67	76	+9 (13.4%)	p = 0.00947
 Chlorophyll Index	33.1	35.4	+2.3 (6.9%)	p = 0.00578
 Biomass %N (C:N analysis)	1.944%	2.286%	+17.6%	n too small*
 CSNT (Cornstalk Nitrate)	79 ppm	938 ppm	+859 ppm	n = 1

* C:N sample sizes are too small for reliable hypothesis testing (n = 3 Control, n = 7 Treatment). Effect sizes are reported instead.

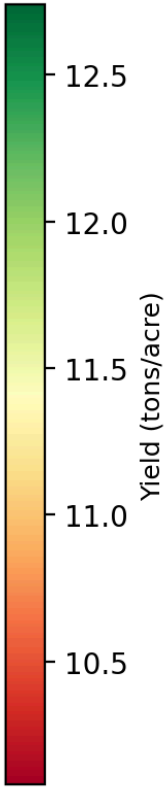
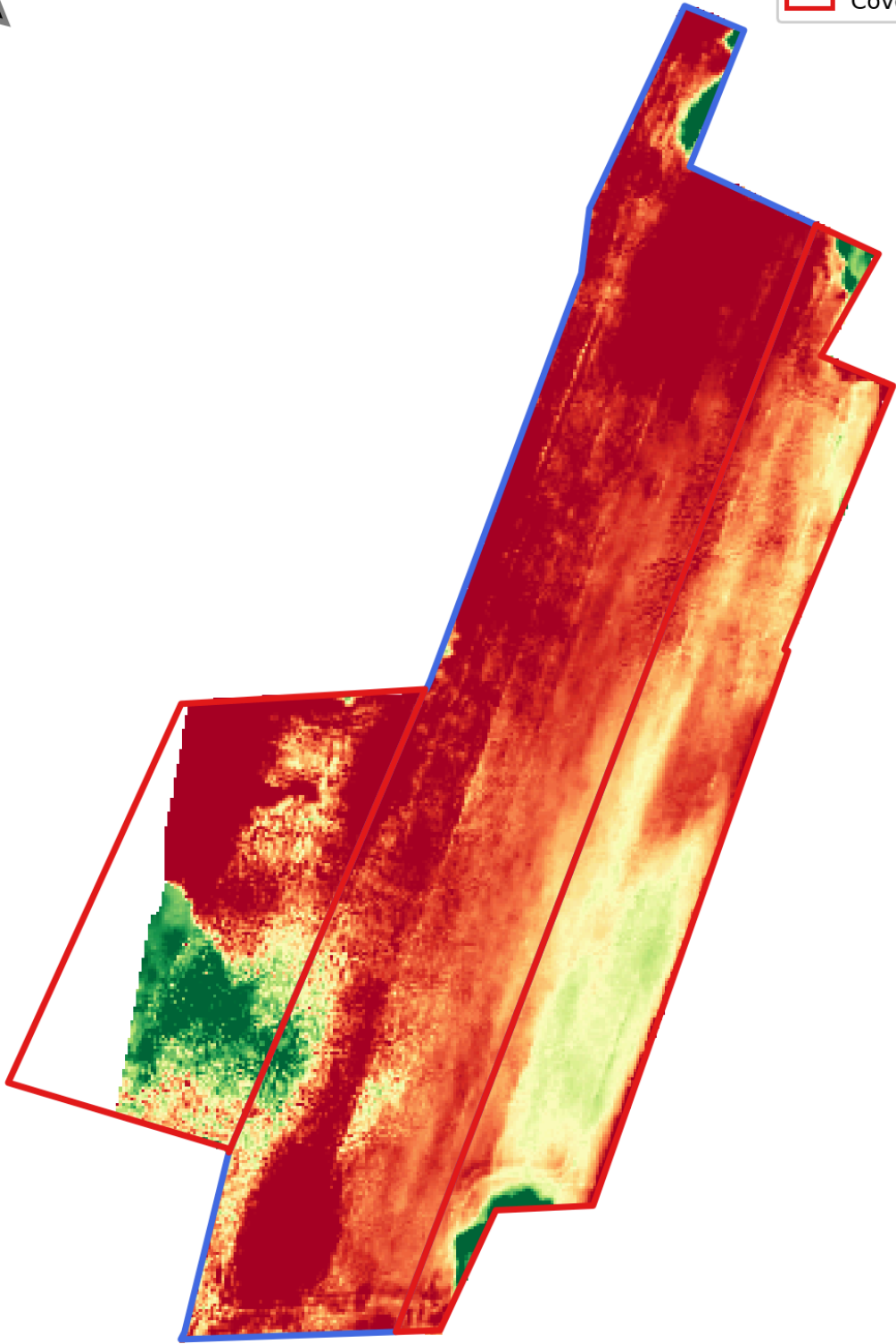
Corn Yield

The map below shows the **corn yield spatial distribution** across Field 3, displayed in tons per acre.

Villnave — Yield



- Villnave
- Control
 - Cover Crop



75 m

Figure 2. Corn yield map.

Nitrogen Balance Index (NBI)

The **Nitrogen Balance Index (NBI)** is measured with the Dualex sensor and reflects the ratio of chlorophyll to flavonoids in the leaf. A higher NBI indicates better nitrogen status, the plant has enough N to build chlorophyll without accumulating excess flavonoids as a stress response. We took two readings per plant on 175 plants (122 in the cover crop strips, 53 in the control), averaged them, and compared the two groups with a two-sample t-test. The difference was statistically significant, meaning it is unlikely to be explained by random variation between plants.

TREATMENT

76 ↑ +9 vs control (13.4%)

Control mean: 67

EFFECT SIZE

13.4%

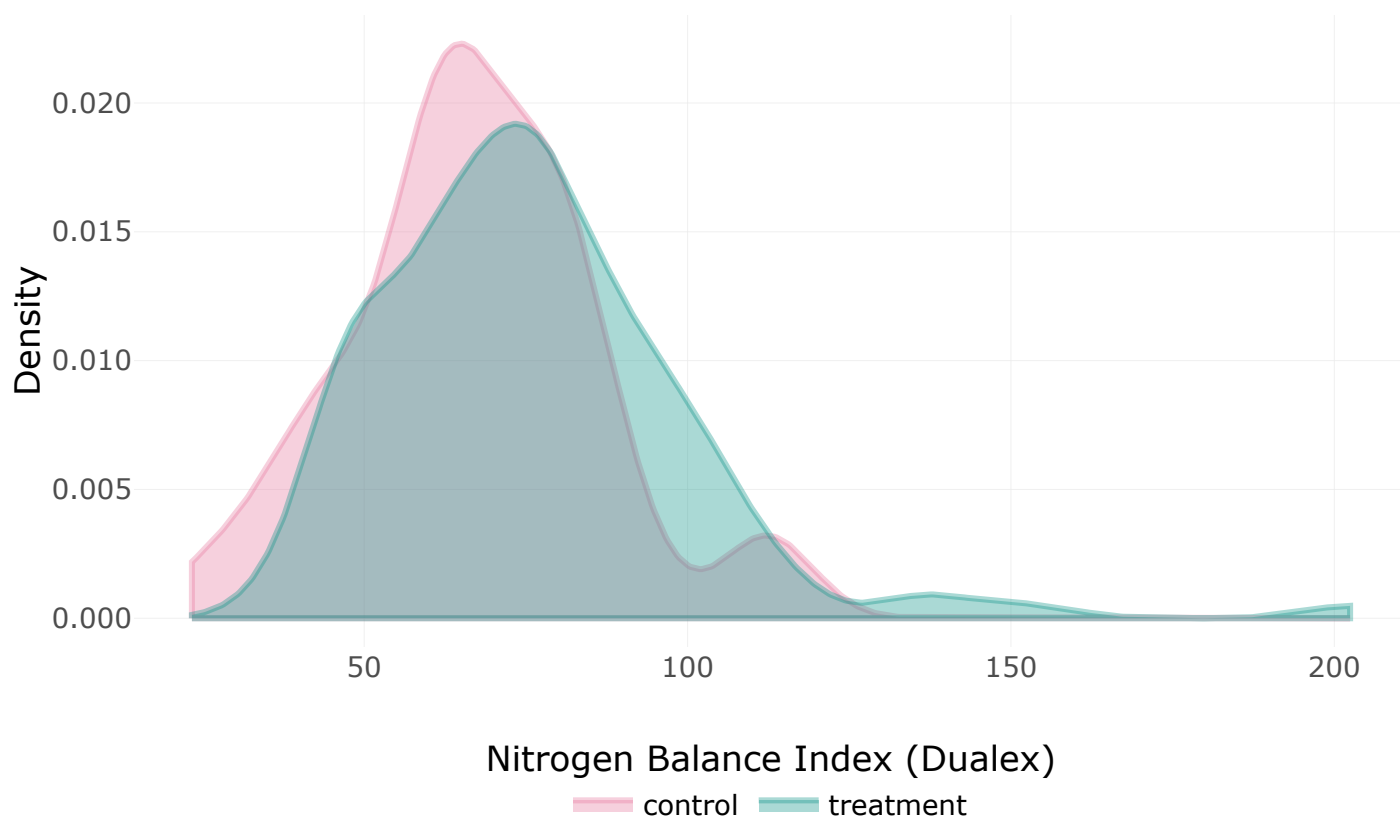
relative difference treatment vs control

p = 0.00947 ✓

Density

Boxplot

Nitrogen Balance Index (NBI) — Plant Average



Chlorophyll (Chl)

The **Chlorophyll Index (Chl)** is also measured with the Dualex sensor and estimates the amount of chlorophyll in the leaf. Chlorophyll is directly linked to the plant's capacity for photosynthesis and is a reliable proxy for nitrogen sufficiency. Higher values indicate greener, more nitrogen-sufficient leaves. As with NBI, two leaves per plant were averaged and treatments compared with a t-test.

TREATMENT

35.4 ↑ +2.3 vs control (6.9%)

Control mean: 33.1

EFFECT SIZE

6.9%

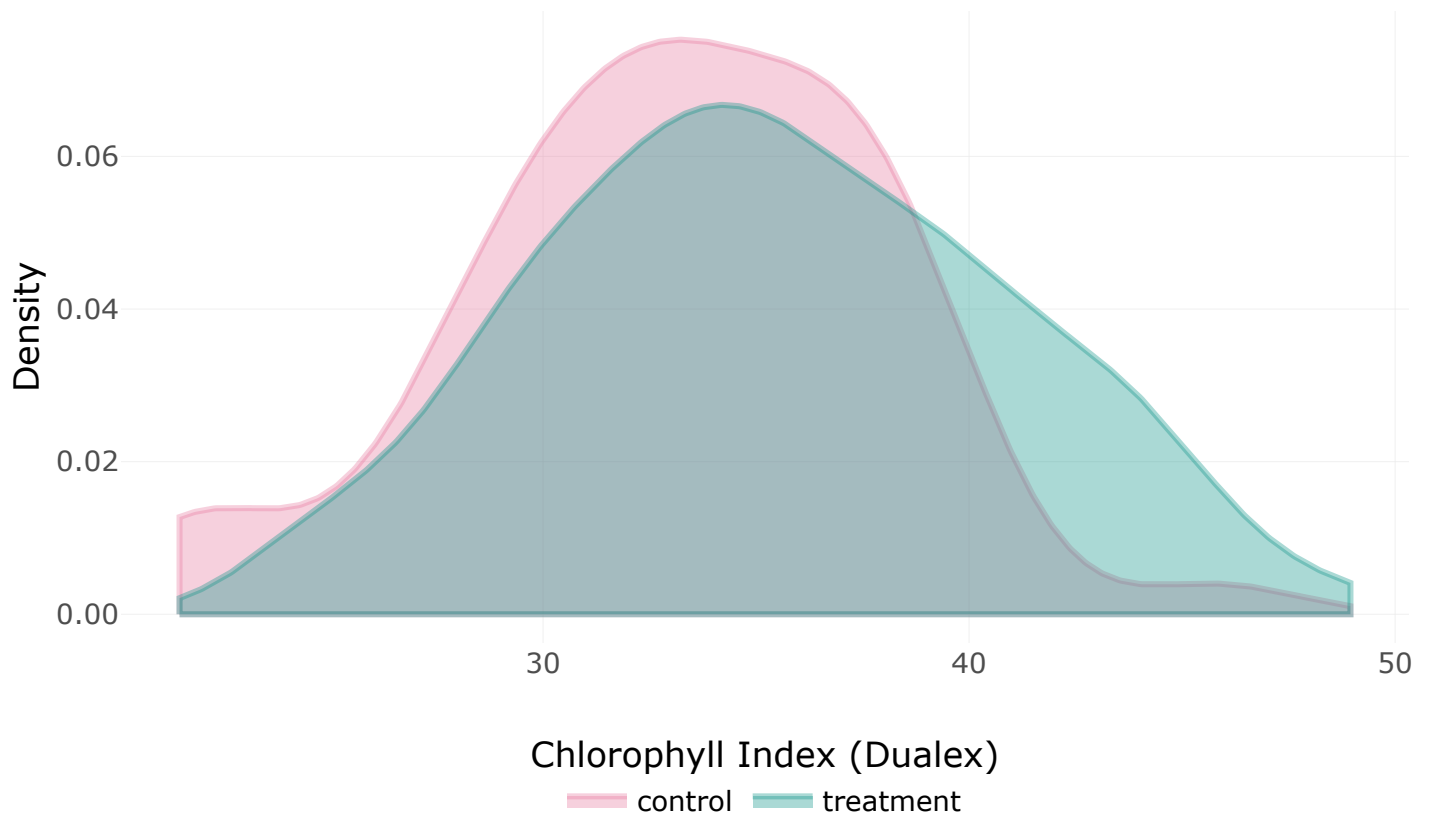
relative difference treatment vs control

p = 0.00578 ✓

Density

Boxplot

Chlorophyll (Chl) — Plant Average



C:N in Corn Biomass

Corn biomass was collected and sent for laboratory C:N analysis. This measures the actual nitrogen concentration in the plant tissue, a direct indicator of how much N the plant was able to take up. **Note:** We only collected a few biomass samples (3 control, 7 treatment), which is too few for a reliable statistical test. Instead, we report an “effect size” (Cohen’s d). It answers a simple question: is the gap between the cover crop and control strips bigger than the normal differences you see from one sample to the next within the same strip? A value near 0 means the two strips look about the same. A value above 0.8 means the gap clearly stands out from that normal variation. With so few samples, treat it as a hint rather than proof.

Total Nitrogen in Biomass (%N)

TREATMENT — MEAN %N

2.286 ↑ +17.6% vs Control

Control mean: 1.944

COHEN'S D (EFFECT SIZE)

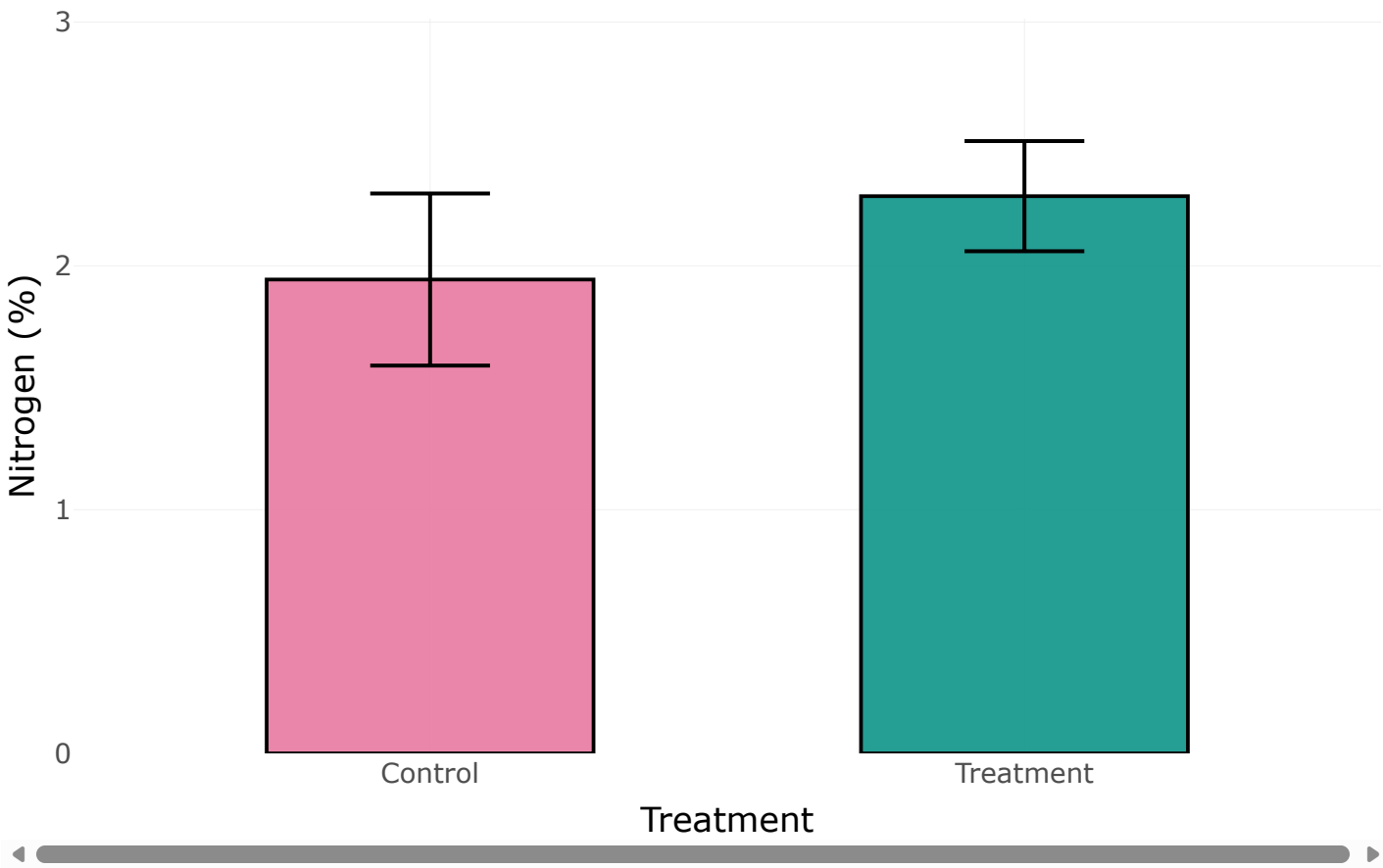
0.57

n = 3 Control · n = 7 Treatment

Bar Chart (%N)

Raw Data (%N)

Nitrogen in Corn Biomass (Mean %N ± SE)



C:N Ratio

TREATMENT — MEAN C:N RATIO

20.88 ↓ -16.5% vs Control

Control mean: 25

COHEN'S D (EFFECT SIZE)

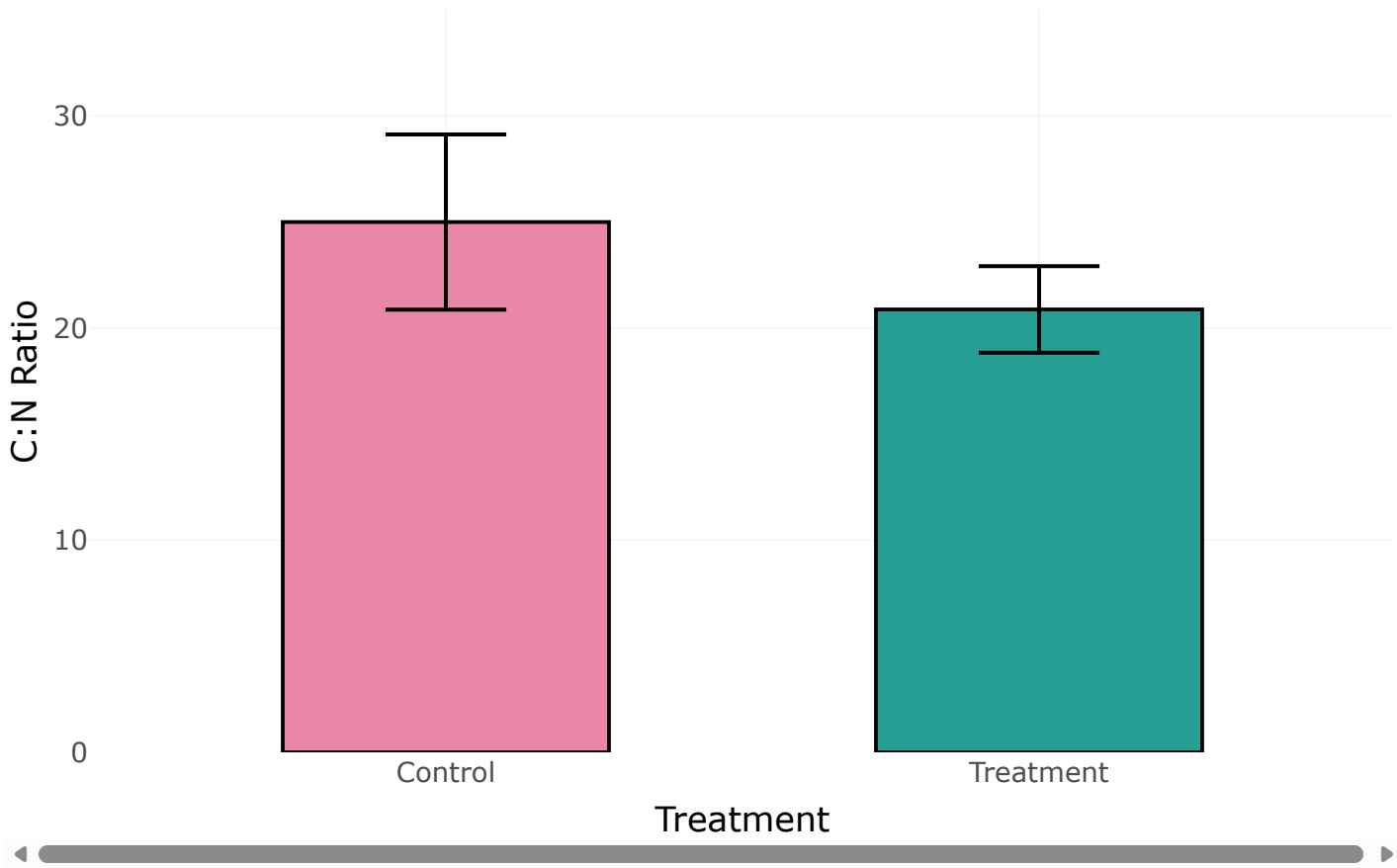
0.7

Lower C:N = more nitrogen-rich tissue

Bar Chart (C:N)

Raw Data (C:N)

C:N Ratio in Corn Biomass (Mean ± SE)



Cornstalk Nitrate Test (CSNT)

The Cornstalk Nitrate Test (CSNT) measures nitrate concentration in the lower cornstalk at the end of the season. It is a post-harvest diagnostic tool: values **below 250 ppm indicate insufficient nitrogen**, while the **optimal range is between 750 and 2000 ppm**. Because only **one sample per treatment** was collected, no statistical comparison is possible, this is a field-level observation.

CONTROL — CSNT

79 ppm

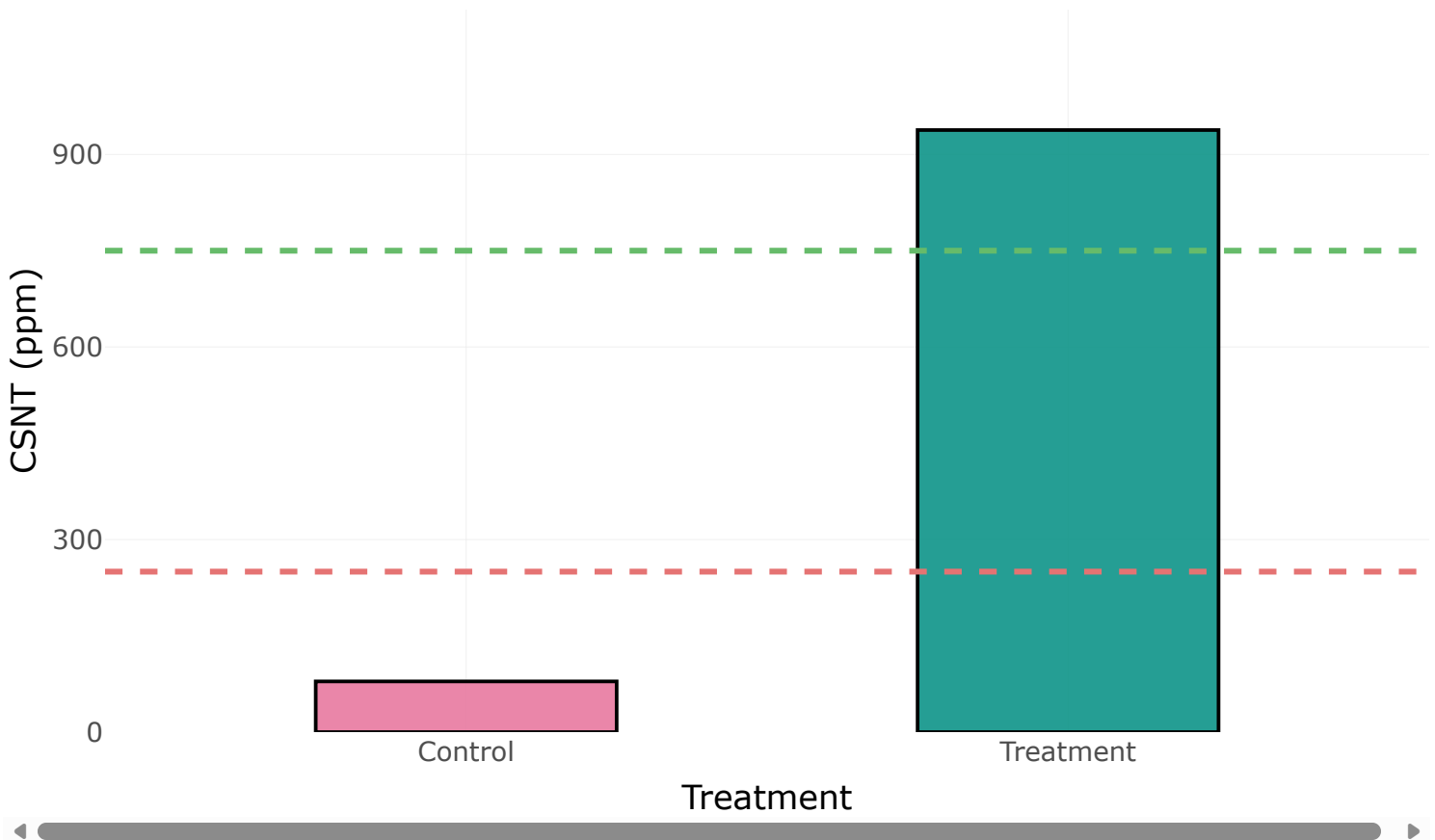
Threshold for sufficient N: 250 ppm

TREATMENT — CSNT

938 ppm

n = 1 per treatment — descriptive only

Cornstalk Nitrate Test (CSNT)



Soil Health Assessment

Soil health was assessed using the **Cornell Soil Health Test** (sampled June 11, 2025, Field 3). The overall quality score was **69/100 (High)**. Chemical indicators are strong: pH (6.3), phosphorus and potassium all scored 98 or higher. On the physical side, water holding capacity (87) and surface hardness (75) came back well, while aggregate stability (55) and subsurface hardness (59) are in the Medium range. The area to watch is biological activity: soil respiration scored 16 (Very Low), and active carbon and soil protein both scored 34 (Low), which means the microbial community that breaks down organic matter and cycles nutrients isn't as active as it could be. Cover cropping works on exactly this: it feeds soil microbes and adds carbon, which helps build biological activity over time.

Comprehensive Assessment of Soil Health

From the Cornell Soil Health Laboratory, Department of Soil and Crop Sciences
School of Integrative Plant Science, Cornell University, Ithaca, NY 14853
<https://soilhealthlab.cals.cornell.edu>



Agricultural Service Provider:
Alex Yole
ay327@cornell.edu

Sample ID: CC35
Field ID: Vilnare field 3
Date Sampled: 06/11/2025
Tillage: no till
Coordinates: Latitude: 42.693276
Longitude: -76.16735

Measured Soil Textural Class: **silt loam**
Sand: **30%** - Silt: **54%** - Clay: **15%**

Group	Indicator	Value	Rating	Constraints
physical	<u>Predicted</u> Available Water Capacity	0.23	87	
physical	Surface Hardness	107	75	
physical	Subsurface Hardness	276	59	
physical	Aggregate Stability	33.5	55	
biological	Organic Matter Soil Organic Carbon: 1.75 / Total Carbon: 1.79 / Total Nitrogen: 0.18	3.4	68	
biological	ACE Soil Protein Index	5.1	34	
biological	Soil Respiration	0.3	16	Soil Microbial Abundance and Activity
biological	Active Carbon	426	34	
chemical	Soil pH	6.3	98	
chemical	Extractable Phosphorus	10.5	100	
chemical	Extractable Potassium	171.8	100	
chemical	Additional Nutrients Ca: 1275.3 / Mg: 147.8 / S: 4.9 Al: 29.6 / B: 0.09 / Cu: 0.15 Fe: 2.2 / Mn: 2.7 / Zn: 0.3		100	

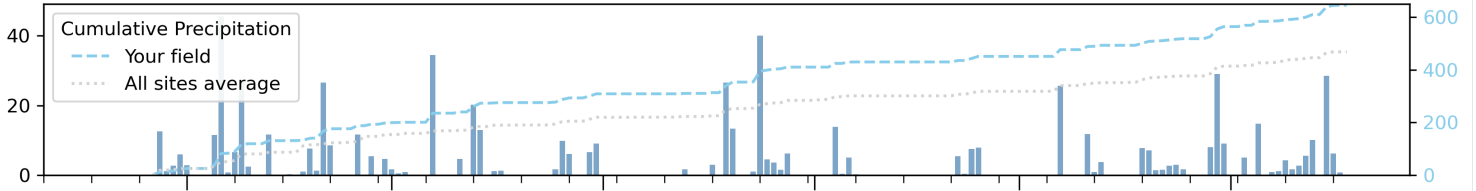
Overall Quality Score: **69** / High

[Open Full Soil Health Report \(PDF\)](#)

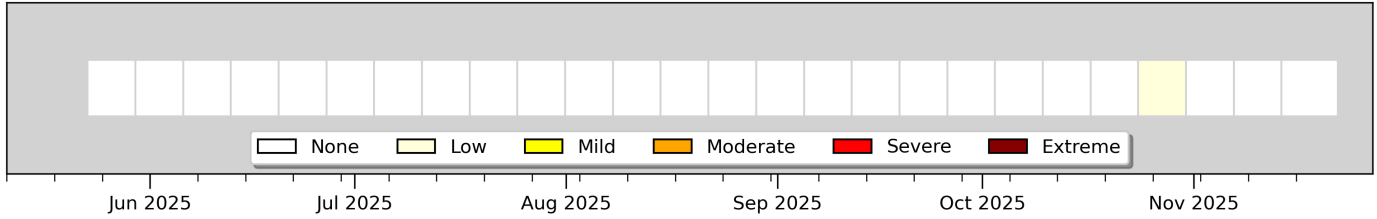
Weather & Drought Conditions

Cumulative precipitation at your field compared to the all-site average, alongside the weekly drought index, from late May through mid-November 2025. Your field had a wetter season than the average site: cumulative rainfall reached about 650 mm (about 26 inches), well above the all-site average of about 470 mm. Rain was frequent in June and July, followed by a mostly dry stretch in the first half of August and several large events in late August. The weekly drought index stayed at None for almost the whole period, with only a brief Low signal in late October.

Local vs All Site Average Precipitation 2025 (in mm)



Weekly Drought Index



Conclusions

LEAF NITROGEN STATUS (NBI)

Cover crop strips showed a **13.4% higher NBI** compared to the control (76 vs 67). Corn plants in the cover crop strips had better nitrogen status mid-season, and the difference was statistically significant.

CHLOROPHYLL INDEX (CHL)

Chlorophyll values were also higher in cover crop strips (**+6.9%**; 35.4 vs 33.1), consistent with the NBI result. Higher chlorophyll reflects greener, more nitrogen-sufficient leaves.

BIOMASS NITROGEN (%N)

Laboratory analysis of corn biomass showed treatment plants had **+17.6% more nitrogen** in their tissue on average (2.286% vs 1.944%). The effect size (Cohen's $d = 0.57$) is moderate, but with only 3 control and 7 treatment samples, and values that overlap between the two groups, this difference could be due to chance.

END-OF-SEASON CSNT

The control stalk sample came in at **79 ppm**, below the 250 ppm sufficiency threshold, while the cover crop sample came in at **938 ppm**, within the optimal range (750–2,000 ppm). This is a large gap and it points in the same direction as the leaf readings, but it comes from a single sample per treatment, so it cannot be tested statistically and should be confirmed with more samples next season.

SOIL HEALTH

Your overall soil health score of **69/100 (High)** reflects good water holding capacity and nutrient levels, with low biological activity: soil respiration (score 16), active carbon (score 34) and soil protein (score 34). Cover cropping adds organic inputs that feed soil microbes, so continuing it is a reasonable way to work on these indicators over time.

